Hermetic-Sheath Toroidal Axion Pickup: Boundary-Value Framework and Optimization Equations

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1 Setup and figure of merit

Lumped-element axion haloscopes operate in the magnetoquasistatic regime where the axion Compton wavelength is much larger than the detector [1–5]. The scan-rate figure of merit, in the resonator-noise-limited regime, is

$$Q \;\equiv\; B_{\max}^4 \, V_{\text{eff}}^4 \,/\, L_p^2, \tag{1}$$

with $B_{\max}$ the peak DC field, L_p the pickup inductance, and V_{eff} the effective volume. The cost is $C \equiv \int B^2 \, dV$ over the high-field magnet bore.

Two geometries are compared:

- **Toroid + hermetic sheath pickup.** Toroidal magnet bore $V_{\text{mag}} : R_t < s < R_t(1+\beta),\ 0 < z < h \equiv \alpha R_t$, with the wound-toroid field $\vec{B}_0 = B_{\max}\, R_t/s \;\hat{\varphi}$ inside V_{mag}. The pickup is a superconducting hermetic sheath surface S_{sc} enclosing V_{mag}, with a slit and external readout path included as part of the physical geometry. Throughout, $\alpha \equiv h/R_t$ and $\beta \equiv a/R_t$ where $a = \beta R_t$ is the radial annulus width.
- **Coaxial-pickup solenoid ("Core").** Solenoidal bore $s < R_m$, length h, with uniform $\vec{B} = B_{\max}\hat{z}$. Pickup is a coaxial transmission line in the field-free annulus $R_m < s < R_{\text{pu}}$, $0 < z < h$, shorted at one end. Closely related geometries appear in DMRadio-m^3 [6, 7] and the GUT-scale proposal [8].

2 Unit-current magnetostatic boundary-value problem

For the sheath pickup, L_p and V_{eff} are both functionals of the same magnetostatic solution: the response of the pickup/readout circuit to one ampere driven through the external terminals. Let Ω be the vacuum domain outside the superconducting material (including the toroidal bore, central hole, slit region, exterior, and the space occupied by the external readout). Define $\vec{A}_1$ and $\vec{B}_1 = \nabla \times \vec{A}_1$ as the fields produced by unit current in the closed pickup/readout loop. The defining problem is

$$\nabla \times (\mu_0^{-1} \nabla \times \vec{A}_1) = \vec{J}_1 \qquad \text{in } \Omega, \tag{2a}$$

$$\nabla \cdot \vec{A}_1 = 0, \tag{2b}$$

$$\hat{n} \cdot \vec{B}_1 = 0 \qquad \text{on the sheath surface } S_{\text{sc}}, \tag{2c}$$

$$\vec{A}_1 \to 0 \qquad \text{as } |\vec{r}| \to \infty, \tag{2d}$$

where $\vec{J}_1$ is the unit-current source through the readout terminals (slit + external return path). The slit and the external return are part of the geometry, not optional corrections: they break the axisymmetry of the closed sheath and are what make a nonzero signal-coupled response possible at all (Sec. 4).

Inductance. From the unit-current solution,

$$\boxed{L_p \;=\; \frac{1}{\mu_0}\int_\Omega |\vec{B}_1|^2\, d^3x.} \tag{3}$$

This is the magnetic energy of the complete pickup/readout circuit; it automatically includes the self-energy of the external return path. If the solution is computed at current I rather than unit current, replace by $L_p = (\mu_0 I^2)^{-1}\int |\vec{B}_I|^2\, d^3x$.

Effective volume. The same $\vec{A}_1$ determines the effective volume by reciprocity with the DC magnet field $\vec{B}_0$:

$$\boxed{V_{\text{eff}} \;=\; \frac{1}{\mu_0 B_{\max}}\int_{V_{\text{mag}}} \vec{B}_0\cdot\vec{A}_1\, d^3x \;=\; \frac{1}{\mu_0}\int_{V_{\text{mag}}} \frac{R_t}{s}\, A_{1,\varphi}(s,\varphi,z)\, d^3x.} \tag{4}$$

Tying V_{eff} and L_p to the *same* $\vec{A}_1$ is essential: combining a V_{eff} computed for one circuit topology with an L_p computed for a different one (e.g. pairing the linked-mode inductance of Sec. 5 with an ABRA-style central-hole loop V_{eff}) is not a consistent calculation.

3 Equivalent surface-current formulation

For a thin ideal sheath the same problem can be cast as an integral equation for the induced surface-current density $\vec{K}_1$ on S_{sc}. Writing

$$\vec{A}_1(\vec{r}) \;=\; \frac{\mu_0}{4\pi}\left[\int_{S_{\text{sc}}} \frac{\vec{K}_1(\vec{r}\,')}{|\vec{r}-\vec{r}\,'|}\, dS' \;+\; \int_{C_{\text{ext}}} \frac{d\vec{\ell}\,'}{|\vec{r}-\vec{r}\,'|}\right], \tag{5}$$

the constraints become

$$\nabla_S\cdot\vec{K}_1 = 0 \qquad \text{away from slit terminals,} \tag{6}$$

$$\int_\gamma \vec{K}_1\cdot\vec{m}\, d\ell = 1 \qquad \text{for any sheath cut } \gamma \text{ crossed once by the readout,} \tag{7}$$

$$\hat{n}\cdot(\nabla\times\vec{A}_1) = 0 \qquad \text{on } S_{\text{sc}}, \tag{8}$$

and the inductance is

$$L_p \;=\; \frac{\mu_0}{4\pi}\left[\int_{S_{\text{sc}}}\int_{S_{\text{sc}}} \frac{\vec{K}_1(\vec{r})\cdot\vec{K}_1(\vec{r}\,')}{|\vec{r}-\vec{r}\,'|}\, dS\, dS' + 2\int_{S_{\text{sc}}}\int_{C_{\text{ext}}} \frac{\vec{K}_1(\vec{r})\cdot d\vec{\ell}\,'}{|\vec{r}-\vec{r}\,'|}\, dS + \int_{C_{\text{ext}}}\int_{C_{\text{ext}}} \frac{d\vec{\ell}\cdot d\vec{\ell}\,'}{|\vec{r}-\vec{r}\,'|}\right], \tag{9}$$

with the standard finite-wire-radius regularization on the external-lead self-term. For a thin ideal sheath this boundary-integral form is often the most direct numerical route.

4 Axisymmetric no-go for the signal-coupled mode

A useful sanity check on any analytic ansatz is the following: an axisymmetric, current-free poloidal field inside a perfectly hermetic toroidal cavity vanishes. Writing $\vec{B}_{\text{pol}} = B_s(s,z)\hat{s} + B_z(s,z)\hat{z}$ and introducing the Stokes stream function ψ with

$$B_s = -\frac{1}{s}\frac{\partial\psi}{\partial z}, \qquad B_z = \frac{1}{s}\frac{\partial\psi}{\partial s}, \tag{10}$$

the current-free condition $\nabla \times \vec{B}_{\text{pol}} = 0$ gives the elliptic equation

$$\psi_{ss} - \frac{1}{s}\psi_s + \psi_{zz} \;=\; 0 \qquad \text{in } V_{\text{mag}}. \tag{11}$$

The Meissner condition $\vec{B}_{\text{pol}} \cdot \hat{n} = 0$ on each of the four sheath faces makes ψ constant on the connected rectangular boundary of the cross-section. By uniqueness for this elliptic Dirichlet problem,

$$\psi \;=\; \text{const}, \qquad \vec{B}_{\text{pol}} \;=\; 0 \qquad \text{everywhere in } V_{\text{mag}}. \tag{12}$$

A uniform axial B_z, or indeed any nonzero axisymmetric poloidal interior field, is *not* a valid leading-order solution for the hermetic problem. Any nonzero signal-coupled response in this geometry requires the slit and external return to be treated as a genuine three-dimensional perturbation that breaks the axisymmetry; there is no axisymmetric closed-form $L_p(\alpha,\beta)$ or $V_{\text{eff}}(\alpha,\beta)$ for the zero-linking signal mode.

5 The one analytic closed-form mode

There is one analytic toroidal mode, but it is signal-blind. If the closed readout circuit links the major-axis cycle once (i.e. the readout wire returns through the central hole), Ampère's law applied to any major-axis loop inside V_{mag} gives

$$\vec{B}_1 \;=\; \frac{\mu_0 I}{2\pi s}\,\hat{\varphi} \qquad \text{inside } V_{\text{mag}}, \tag{13}$$

with the corresponding inductance

$$L_{\text{linked}} \;=\; \frac{\mu_0 h}{2\pi}\,\ln(1+\beta) \;=\; \frac{\mu_0 \alpha R_t}{2\pi}\,\ln(1+\beta). \tag{14}$$

For ABRA-10cm parameters $(R_t,\alpha,\beta) = (3\,\text{cm}, 4, 1)$, this gives $L_{\text{linked}} \approx 16.6\,\text{nH}$.

The reciprocity overlap with the wound-toroid axion source vanishes for this mode: a vector potential generating the linked toroidal field $\vec{B} = (\mu_0 I/2\pi s)\hat{\varphi}$ can be chosen with no $\hat{\varphi}$ component in V_{mag} (e.g. $\vec{A} = (\mu_0 I/2\pi)\ln(s/s_0)\,\hat{z}$), so $\vec{B}_0 \cdot \vec{A}_1 = 0$ pointwise and

$$V_{\text{eff}}^{\text{linked}} \;=\; 0. \tag{15}$$

L_{linked} is a real inductance for this mode and is useful as a cross-check for any 3D code, but it must not be paired with a nonzero central-hole V_{eff} from a different (loop-pickup) mode.

6 Core: long-solenoid coaxial pickup

For the Core geometry, with the long-solenoid limit $h \geq R_m$ and $r \equiv R_{\rm pu}/R_m$, the standard formulas are analytic:

$$V_{\rm eff}^{\rm core} \;=\; \tfrac{1}{2}\,h\,R_m^2\,\ln r, \qquad L_p^{\rm core} \;=\; \frac{\mu_0 h}{2\pi}\,\ln r. \tag{16}$$

The Core cost and bounding volume are

$$C^{\rm core} \;=\; \pi R_m^2 h\, B_{\max}^2, \qquad V_{\rm tot}^{\rm core} \;=\; \pi R_{\rm pu}^2 h \;=\; \pi r^2 R_m^2 h. \tag{17}$$

At fixed bounding volume with $h = R_m$ saturating the long-solenoid constraint,

$$Q^{\rm core}\big|_{V_{\rm tot}} \;=\; \frac{B_{\max}^4\, V_{\rm tot}^{10/3}}{4\,\pi^{4/3}\mu_0^2}\,\frac{\ln^2 r}{r^{20/3}}, \qquad r^* = e^{3/10} \approx 1.350. \tag{18}$$

At fixed magnet cost (just the bore), R_m is fixed by C and $Q^{\rm core}|_C \propto \ln^2 r$ is unbounded in $R_{\rm pu}$; a finite Core optimum at fixed cost requires an envelope-dependent cost term beyond $\int B^2\,dV$.

7 Toroid optimization equations

After the 3D unit-current problem (2) is solved for a specified slit/readout geometry, define dimensionless functions

$$\ell(\alpha,\beta;\mathcal{G}) \;=\; \frac{L_p}{\mu_0 R_t}, \qquad v(\alpha,\beta;\mathcal{G}) \;=\; \frac{V_{\rm eff}}{R_t^3}, \tag{19}$$

where $\mathcal{G}$ encodes the non-universal geometric data: slit width and orientation, external return path, shielding, lead placement, and any ports or finite conductor dimensions. The Q figure of merit is then

$$Q^{\rm tor} \;=\; B_{\max}^4\,\frac{R_t^{10}}{\mu_0^2}\,\frac{v(\alpha,\beta;\mathcal{G})^4}{\ell(\alpha,\beta;\mathcal{G})^2}. \tag{20}$$

The toroidal magnetic-energy cost remains analytic,

$$C^{\rm tor} \;=\; \int_{V_{\rm mag}} B_0^2\,dV \;=\; 2\pi\,B_{\max}^2\,R_t^3\,\alpha\,\ln(1+\beta), \tag{21}$$

as does the bounding volume $V_{\rm tot}^{\rm tor} = \pi R_t^3 \alpha (1+\beta)^2$. Eliminating R_t gives, at fixed bounding volume,

$$Q^{\rm tor}\big|_{V_{\rm tot}} \;=\; \frac{B_{\max}^4\,V_{\rm tot}^{10/3}}{\mu_0^2\,\pi^{10/3}}\;\frac{v(\alpha,\beta;\mathcal{G})^4}{\ell(\alpha,\beta;\mathcal{G})^2\,\alpha^{10/3}(1+\beta)^{20/3}}, \tag{22}$$

and at fixed magnetic cost,

$$Q^{\rm tor}\big|_{C} \;=\; \frac{B_{\max}^4}{\mu_0^2}\,\left[\frac{C}{2\pi B_{\max}^2\,\alpha\,\ln(1+\beta)}\right]^{10/3}\frac{v(\alpha,\beta;\mathcal{G})^4}{\ell(\alpha,\beta;\mathcal{G})^2}. \tag{23}$$

These are the correct optimization equations. The functions ℓ and v must come from the *same* 3D unit-current solution for the *same* $\mathcal{G}$; they are not universal functions of α and β alone.

8 What is analytic and what requires FEM

Analytic, valid:

- The toroidal magnet cost (21) and bounding volume.
- The Core pickup formulas (16), in their long-solenoid regime.
- The linked wound-toroid inductance (14) and the corresponding $V_{\text{eff}}^{\text{linked}} = 0$.

Requires 3D FEM (or boundary-integral solution):

- L_p for the hermetic sheath pickup with a specified slit/readout geometry.
- V_{eff} for the same pickup, computed from the same $\vec{A}_1$ used for L_p.
- Any toroid/Core comparison (ratio at fixed V_{tot} or fixed cost). With ℓ and v known for a chosen $\mathcal{G}$, the optimization equations above give the comparison; without them, no ratio is reliable.

Cross-checks the FEM should satisfy:

- In the zero-slit limit, the axisymmetric no-go (Sec. 4) requires the signal-coupled response to vanish, $V_{\text{eff}} \to 0$.
- In the limit where the external return is routed through the central hole, the code should recover $L_p \to L_{\text{linked}}$ from (14) and a V_{eff} that, for the ideal axisymmetric wound-toroid source, becomes parametrically small in proportion to any small A_φ admitted by the gauge / boundary conditions.
- The reciprocity expression (4) should agree with a direct Faraday-law calculation $V_{\text{eff}} = -(1/B_{\max}) \int (\partial_t)^{-1} \mathcal{E}\, dt$ for a test modulation of the axion source.

9 Bottom line

The optimization framework reduces to a single, well-posed electromagnetic calculation: solve (2) once for the chosen pickup/readout geometry; read off L_p from (3) and V_{eff} from (4); plug into the dimensionless optimization equations. No closed-form analytic $L_p(\alpha, \beta)$ or $V_{\text{eff}}(\alpha, \beta)$ exists for the zero-linking signal-coupled hermetic-sheath mode, and any such formula offered without an underlying 3D solution should be treated as a topology-specific heuristic and explicitly checked against the Meissner boundary conditions before use. With $\ell(\alpha, \beta; \mathcal{G})$ and $v(\alpha, \beta; \mathcal{G})$ in hand, the comparison to the Core is straightforward and the design optimization is decoupled into a one-time magnetostatic computation per sheath topology and a downstream analytic optimization in (α, β).

References

[1] P. Sikivie, N. Sullivan, and D. B. Tanner. Proposal for Axion Dark Matter Detection Using an LC Circuit. *Phys. Rev. Lett.*, 112(13):131301, 2014.

[2] Yonatan Kahn, Benjamin R. Safdi, and Jesse Thaler. Broadband and Resonant Approaches to Axion Dark Matter Detection. *Phys. Rev. Lett.*, 117(14):141801, 2016.

[3] Saptarshi Chaudhuri, Kent Irwin, Peter W. Graham, and Jeremy Mardon. Optimal Impedance Matching and Quantum Limits of Electromagnetic Axion and Hidden-Photon Dark Matter Searches. 3 2018.

[4] Saptarshi Chaudhuri, Kent D. Irwin, Peter W. Graham, and Jeremy Mardon. Optimal Electromagnetic Searches for Axion and Hidden-Photon Dark Matter. 4 2019.

[5] Joshua N. Benabou, Joshua W. Foster, Yonatan Kahn, Benjamin R. Safdi, and Chiara P. Salemi. Lumped-element axion dark matter detection beyond the magnetoquasistatic limit. *Phys. Rev. D*, 108(3):035009, 2023.

[6] L. Brouwer et al. Projected sensitivity of DMRadio-m3: A search for the QCD axion below 1 μeV. *Phys. Rev. D*, 106(10):103008, 2022.

[7] A. AlShirawi et al. Electromagnetic modeling and science reach of DMRadio-m^3. 2 2023.

[8] L. Brouwer et al. Proposal for a definitive search for GUT-scale QCD axions. *Phys. Rev. D*, 106(11):112003, 2022.